

Methods Dossier for 50 Years of Symbolic Equation Discovery: Revisiting and Improving BACON against Modern Symbolic Regression

Computer Science Tripos Part II Project Methods Dossier

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0. Executive Summary

We will **rebuild BACON.3**, build **BACON.7 (per Miller)**, and **compare** it to a strong **PySR** baseline across synthetic functions, textbook physics laws, and one small real-world dataset. The comparison is **fair** (shared operator libraries; consistent effort via iteration-parity or wall-clock caps) and **reproducible** (fixed seeds, single-command runners, version pinning). Primary outcomes are (i) exact recovery on clean benchmarks and (ii) reliable test R^2 under matched conditions. **Additional evaluations** (reported separately) may include parsimony measures, extrapolation checks, stability over seeds, and performance profiles.

0.1 Problem Statement & Objectives

Problem. Do invariant-first pipelines (BACON-style) retain advantages in **exactness, speed, and interpretability** on small physics-style problems, compared with direct SR search (PySR)?

Objectives.

1. **Rebuild BACON.3** with transparent decision logging.
2. **Specify and build BACON.7 (per Miller):** BACON.3 layering, layer-methods, symmetry apply, and averaging; no new modules beyond the paper.
3. **Compare** BACON.3/BACON.7 and PySR under matched settings on curated tasks.
4. **Report** exactness and test R^2 as primary metrics; list parsimony, extrapolation, stability, and performance profiles as *additional evaluations*.

0.2 Datasets & Splits

0.2.1 Synthetic

Functions (initial set):

- S-1: $y = x_1 + x_2$
- S-2: $y = x_1 \times x_2$
- S-3: $y = \frac{x_1}{x_2+1}$
- S-4: $y = x_1^2 + x_2^2$

Core head-to-head: S-1...S-4 using the shared operator set $\{+, -, \times, /\}$.

Additional analyses: S-5/S-6 (enable $\sin / \exp$ with domain guards).

Domains: chosen to avoid invalid regions.

Noise: $\sigma \in \{0.0, 0.01, 0.05, 0.1\} \times$ dynamic range.

N: 400–800 per synthetic.

Splits: 75/25 train/test (seeded). $5\times$ bootstrap resamples for stability.

Extrapolation: out-of-range slab (e.g., train $x \in [-3, 3]$, test-extra $x \in [3, 4]$).

0.2.2 Textbook laws

T-1 Ohm: $V = IR$ ($I \in [0, 2]$, $R \in [1, 10]$)

T-2 Hooke: $F = kx$ ($x \in [-1, 1]$)

T-3 Free fall: $s = \frac{1}{2}gt^2$ ($t \in [0, 2]$)

T-4 Ideal Gas (simplified): $PV = nRT$ (n, R constants; vary P, V, T within sensible lab ranges)

T-5 Black's law (Stefan–Boltzmann simplified): $P \propto T^4$

Synthetic sampling with light measurement noise.

0.2.3 Public/realistic

R-1/R-2: 1–2 small, low-dimensional regression sets with interpretable structure (kept modest to fit compute/time). If tight on time, defer to next milestone.

Catalogue. Appendix A enumerates IDs, formulas, ranges, N, noise, extrapolation slabs, and notes. This table is the single source of truth used by the runners.

0.3 Methods

0.3.1 BACON.3 (baseline)

Core idea. Propose invariants (ratios/products/powers) that appear constant or monotone; test; and algebraically reconstruct relations.

Implementation. Deterministic passes over a fixed transform set; median-based robustness where applicable; fixed seeds for any tie-breaking.

Outputs. A symbolic law, residual diagnostics, and a **decision log** (human-readable trace of tests leading to the law).

Limits. Sensitive to noise and nested transcendental structure unless hints are allowed.

0.3.2 Improved BACON — BACON.7 (per Miller)

Definition. The “improved” variant here is **Miller’s BACON.7**: BACON.3 layering plus symmetry reuse and averaging, with layer-method choices and managing-layer parameters ($\varepsilon, \delta, c_val$, scaling) to handle noise and tie-breaking.

Key mechanisms (no new modules beyond Miller):

- **BACON.3 layering:** sequentially add variables via invariant cues.
- **Layer-methods:** selection policies such as `popular` and `min_mse`.
- **Symmetry apply & averaging:** reuse symmetries and average equivalent forms where applicable.
- **Managing-layer knobs:** ε/δ thresholds, `c_val`, and scaling as in the manuscript.

Outputs: symbolic law, constants, residuals, and a readable **decision log**; settings ($\varepsilon, \delta, c_val$, selection method) saved for reproducibility.

Operator note: BACON.7 runs with the same operator set declared for each dataset (see §3.4). For the **core head-to-head**, that set is $\{+, -, \times, /\}$.

0.3.3 PySR (modern SR baseline)

Search. Multi-population evolutionary search with `evolve`→`simplify`→`optimise`; age-regularised replacement; adaptive parsimony.

Constants. Local refits (e.g., BFGS/Optim.jl).

Operator library: Use $\{+, -, \times, /\}$ for all datasets to match BACON.7 capacity.

Constraints. `maxsize`, `maxdepth`.

Exports. SymPy (expression size, LaTeX); optional JAX/PyTorch for checks.

0.4 Experimental Protocol

0.4.1 Hyperparameters

Comparison rule: Use **either** iteration-parity **or** wall-clock caps (or both). Default is **iteration-parity** to mirror Miller; caps act as a safety bound. We record both the effort setting and the observed time.

PySR (starters):

- **population_size:** 1000 ($\times 5$ islands)
- **niterations:** **15** for core head-to-head (parity with BACON’s effort counter), **or** until wall-clock cap (see §4.2)
- **maxsize:** 30 nodes; **maxdepth:** 7
- **parsimony:** adaptive (default)
- **loss:** MSE; **constant_optimisation:** **on**
- Simplify each iteration; early-accept improvements

BACON.7:

- BACON.3 layering with BACON.7 extensions (layer-methods; symmetry apply; averaging)
- Managing-layer params: $\varepsilon/\delta/c_val/scales$ per-dataset config; stop on plateau or pass limit

0.4.2 Budgets, seeds, and logging

- **Wall-clock caps:** 15 min per method per synthetic/textbook; 30 min per public set.
- **Replicates:** 5 seeds per method per dataset.
- **Logs (CSV/JSONL):** `run_id`, `dataset_id`, `method`, `seed`, `time_s`, `expr`, `train_R2`, `test_R2`, `test_extra_R2`, `MAE`, `size_nodes`, `status` (ok/timeout/fail), `notes`.

0.4.3 Metrics & comparisons

Primary (Miller-aligned):

- **Exactness** (synthetics/textbook): algebraic equivalence to ground truth; constants within 3–4 s.f.
- **Accuracy:** test R^2 (primary) + MSE/MAE (robustness).

Additional evaluations (modern SR practice; reported separately):

- **Parsimony** (SymPy node count).
- **Stability** (SD/IQR across seeds).
- **Extrapolation** (R^2 on out-of-range slab).
- **Performance profiles** (AUC over error thresholds across datasets).